Imaging wood for tree-ring measurement

Introductory dendrochronology · Lesson 4 supplement. The Yoshino cedar study describes an imaging method; the koyamaki study illustrates the field context. Together, they show how log-end images support ring-width measurement.



Example of a wood cross-section. Photo: Hitoshi Yonenobu, 2022 teaching material. This is not a photograph of the camera setup or the koyamaki fieldwork. Yoshino cedar study.

Before imaging

Record specimen ID, provenance, the surface, and the condition of the pith and bark side. Select a surface on which boundaries can be read, including the narrowest rings and any cracks or damage.

In the Yoshino study, the relationship between camera-to-surface distance and image resolution was calibrated in advance. A variable-length spacer helped maintain distance. Lens performance and field conditions also affect effective resolution.

From images to measurement and crossdating

1 Record position and scale

Attach a scale and link the specimen ID, image sequence, orientation and proposed measurement path. The scale allows distances in the image to be converted to physical lengths.

2 Capture overlapping close views

Photograph successive sections from the pith side toward the bark side at a resolution that permits boundary identification. In the Yoshino study, adjacent frames typically overlapped by one-half to two-thirds. Rings in the overlap help establish the order and correspondence of images.

3 Add wider views

Wide rings or complex surfaces can make short close-up segments difficult to match. The Yoshino study added wider views when needed. Supplementary Figure S1 of the koyamaki study shows the log yard, cross-section, imaging and a close view of rings.

4 Calibrate, measure and crossdate

Calibrate image distances using the scale, identify boundaries and calculate widths. Compare overlapping frames, other radii and other specimens; return to the original images and records when uncertain. The Yoshino study measured widths from images and averaged radii within specimens where two were available.

The Yoshino imaging protocol targeted approximately 0.01 mm measurement precision; lens performance and field conditions also matter, and an image alone does not guarantee that precision.

References

Yonenobu et al. (2026), Yoshino cedar, supplementary text S2.2, p. 3.

Yonenobu et al. (2026), koyamaki, supplementary text S2.1 and Figure S1, p. 4.